Construction Chemicals

Product Datasheet
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Jointing and Tiling



O-Silica

Silica Fume
for High-Performance Concrete

Product Description

O-Silica is a silica fume (micro-silica) addition for high-performance concrete. It increases compressive and flexural strength, reduces permeability and chloride ingress and improves sulfate resistance.

Uses

- High-performance concrete.
- Precast elements.
- Marine structures.
- Bridges.
- Industrial floors.

Advantages

- Increases strength.
- Reduces permeability.
- Improves chloride and sulfate resistance.
- Dense concrete matrix.
- Dosage 5 – 10 % of cement.

Application

1. **Mix Design**
   Include 5 – 10 % of cement weight in the approved mix design.
2. **Dosing**
   Add with the aggregates and cement at the batching plant.
3. **Mixing**
   Extend mixing time to disperse fully; adjust superplasticiser.

Consumption

5 – 10 %
of cement weight

Packaging

25 kg bags.

Storage & Shelf Life

Store dry in closed bags, protected from moisture.

Technical Data

Typical values at 25 °C

Property	Typical Value	Test Method
SiO_2 content	85 – 92 %	—
Bulk density	500 – 700 kg/m³	—
Specific surface	15,000 – 25,000 m²/kg	—
Moisture	≤ 3 %	—

Consumption

Use	Dosage
High-strength concrete	7 – 10 %
Durability / low permeability	5 – 8 %

Consumption is theoretical and varies with substrate porosity, profile and application method.

Application Conditions

- Use only in an approved mix design.
- Use with a superplasticiser.
- Cure immediately after finishing.

Application Notes

- Use with a superplasticiser.
- Extend mixing time.
- Cure carefully; silica fume concrete is sensitive to early drying.

Health & Safety

- Wear gloves, safety goggles and a dust mask when mixing.
- Contains cement; alkaline when wet. Avoid skin and eye contact.
- Avoid inhaling dust.

First Aid & Cleaning

- Eyes: rinse immediately with plenty of clean water for at least 15 minutes and seek medical advice.
- Skin: wash with soap and water; cement is alkaline when wet.
- Inhalation of dust: move to fresh air.
- Ingestion: rinse mouth and seek medical advice.
- Clean equipment with water.

The information in this datasheet is based on laboratory tests and practical experience and is given in good faith. Users should verify product suitability for their specific application. O MIX reserves the right to update this information without prior notice.